

ELECTRIC VEHICLE ANALYSIS





By **BHARATH**



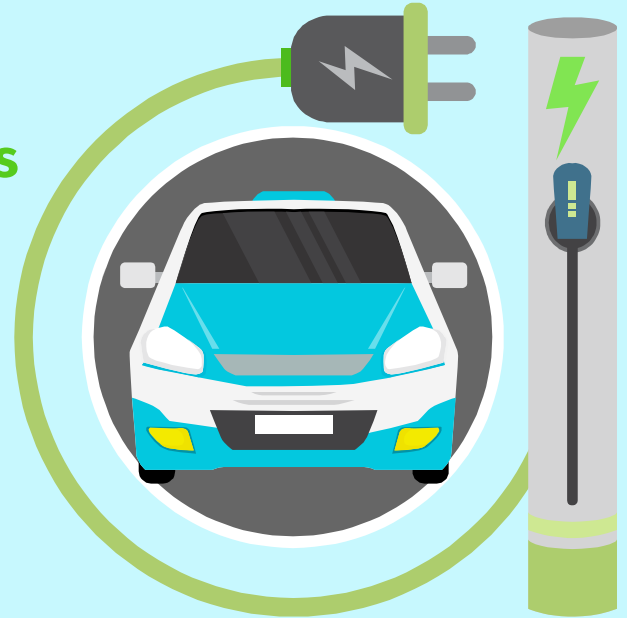
INDIA AT COP-28



50% of energy from **non-fossil-fuels**

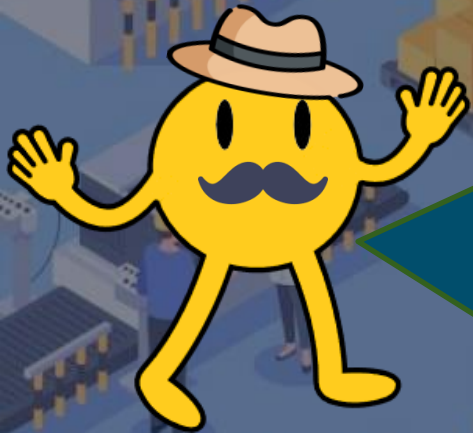


Reduce Co2 emission by **1bn** tons





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BRUCE
HARIYALI



CHIEF
BRUCE HARIYALI

Problem Statement



Atliq Motors is an automotive giant from the USA specializing in electric vehicles (EV).



In the last 5 years, our market share rose to 25% in electric and hybrid vehicles segment in North America.



As a part of our expansion plans, they wanted to launch their bestselling models in India where their market share is less than 2%.



So I wanted to do a detailed market study of existing EV/Hybrid market in India before proceeding further

An isometric illustration of a modern, smart factory. The scene is filled with various industrial elements: robotic arms (yellow and black) are positioned over conveyor belts and workstations, some holding yellow boxes. Workers in blue uniforms and yellow hard hats are engaged in different tasks; some are operating control panels with multiple monitors, while others are inspecting products or using handheld devices. Pallets stacked with yellow boxes are visible throughout the facility. In the background, there are large storage units and more complex machinery. The overall color palette is dominated by blues and greys, with yellow used for accents like the robots and boxes. The perspective is from an elevated angle, looking down into the factory floor.

DATA

01

dim_date



	A	B	C
1	date	fiscal_year	quarter
2	01-Apr-21	2022	Q1
3	01-May-21	2022	Q1
4	01-Jun-21	2022	Q1
5	01-Jul-21	2022	Q2
6	01-Aug-21	2022	Q2

TABLES

02

electric_vehicle_sales_by_state



	A	B	C	D	E
1	date	state	vehicle_category	electric_vehicles_sold	total_vehicles_sold
2	01-Apr-21	Sikkim	2-Wheelers	0	398
3	01-Apr-21	Sikkim	4-Wheelers	0	361
4	01-May-21	Sikkim	2-Wheelers	0	113
5	01-May-21	Sikkim	4-Wheelers	0	98
6	01-Jun-21	Sikkim	2-Wheelers	0	229

03

electric_vehicle_sales_by_maker



	A	B	C	D
1	date	vehicle_category	maker	electric_vehicles_sold
2	01-Apr-21	2-Wheelers	OLA ELECTRIC	0
3	01-Apr-22	2-Wheelers	OKAYA EV	0
4	01-May-21	2-Wheelers	OLA ELECTRIC	0
5	01-Jun-21	2-Wheelers	OLA ELECTRIC	0
6	01-Jul-21	2-Wheelers	OLA ELECTRIC	0

SQL – Key definitions

ENTITY RELATIONSHIP DIAGRAM (ERD) :

It is a visual representation of the structure of a database, showing how different entities (like tables) relate to each other

PRIMARY KEY :

is a column (or a set of columns) in a table that uniquely identifies each row in that table. It ensures that no two rows have the same value in this column, and it cannot contain NULL values. Think of it as a unique identifier for each record.

FOREIGN KEY :

is a column (or a set of columns) in one table that refers to the primary key in another table. It creates a link between the two tables, ensuring data integrity. The foreign key can contain duplicate values and can have NULL values.

STAR SCHEMA:

is a type of database schema that organizes data into fact and dimension tables, resembling a star shape, and is optimized for querying large datasets¹

SQL

Primary Key

dim_date

	A	B	C
1	date	fiscal_year	quarter
2	01-Apr-21	2022	Q1
3	01-May-21	2022	Q1
4	01-Jun-21	2022	Q1
5	01-Jul-21	2022	Q2
6	01-Aug-21	2022	Q2

many-to-one relationships

Foreign Key

Foreign Key

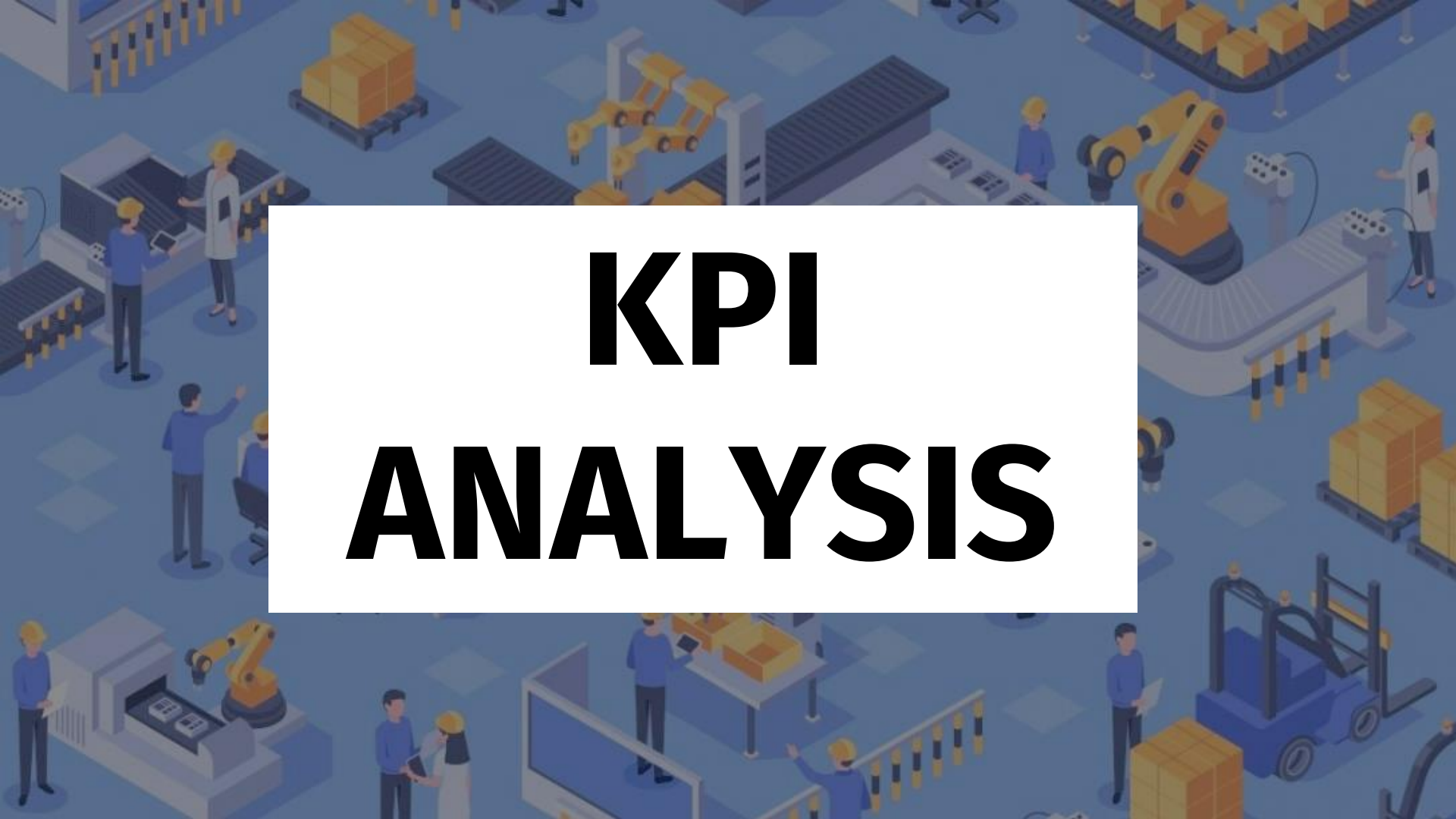
electric_vehicle_sales_by_state

	A	B	C	D
1	date	vehicle_category	maker	electric_vehicles_sold
2	01-Apr-21	2-Wheelers	OLA ELECTRIC	0
3	01-Apr-22	2-Wheelers	OKAYA EV	0
4	01-May-21	2-Wheelers	OLA ELECTRIC	0
5	01-Jun-21	2-Wheelers	OLA ELECTRIC	0
6	01-Jul-21	2-Wheelers	OLA ELECTRIC	0

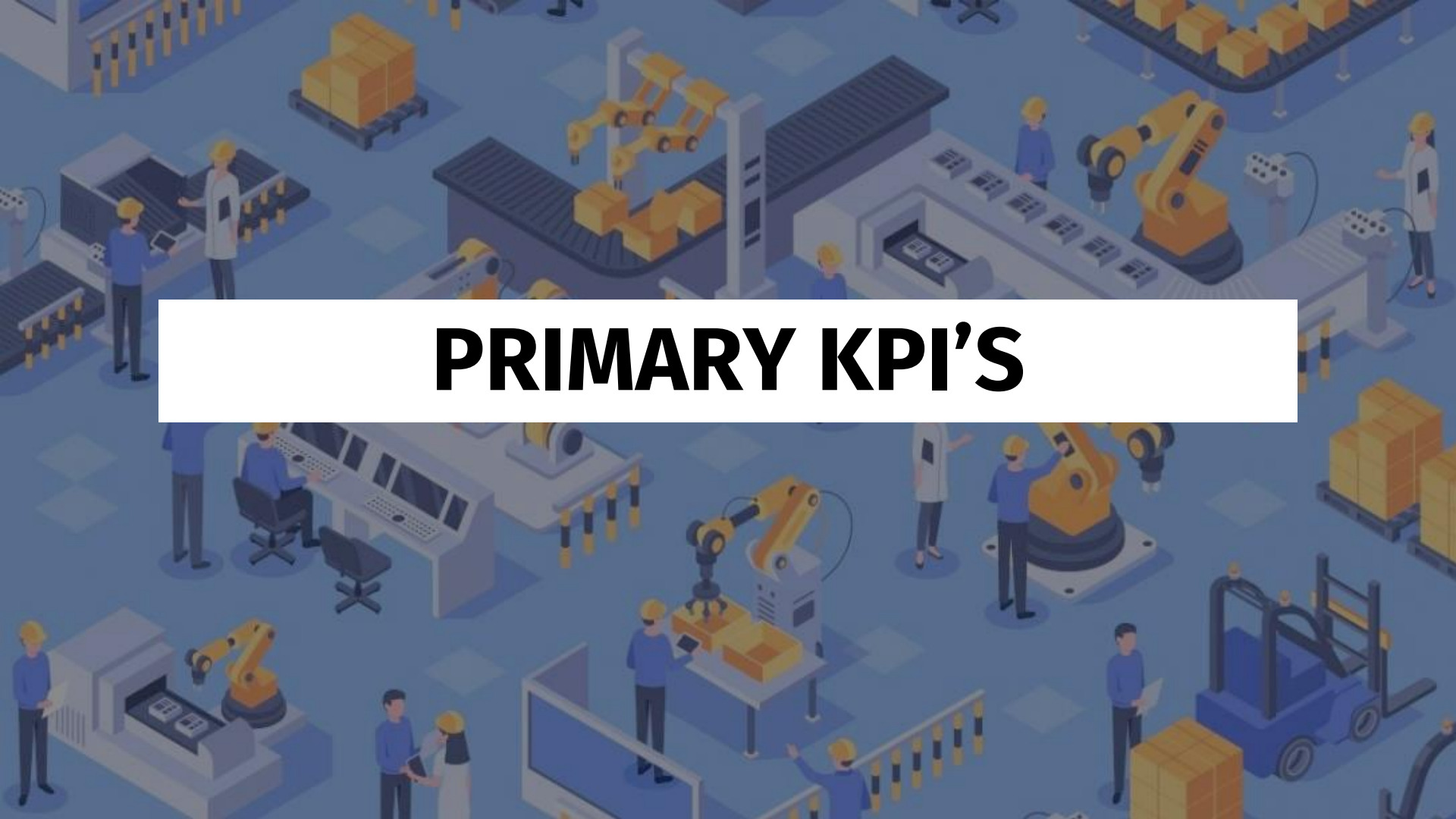
electric_vehicle_sales_by_makers

	A	B	C	D	E
1	date	state	vehicle_category	electric_vehicles_sold	total_vehicles_sold
2	01-Apr-21	Sikkim	2-Wheelers	0	398
3	01-Apr-21	Sikkim	4-Wheelers	0	361
4	01-May-21	Sikkim	2-Wheelers	0	113
5	01-May-21	Sikkim	4-Wheelers	0	98
6	01-Jun-21	Sikkim	2-Wheelers	0	229

STAR SCHEMA

An isometric illustration of a modern factory floor. The scene is filled with various industrial elements: yellow robotic arms are positioned over conveyor belts and workstations; workers in blue uniforms and yellow hard hats are engaged in different tasks, some using tablets or laptops; stacks of yellow boxes are visible on pallets and in storage areas; and a blue forklift is parked near some of the boxes. The overall color palette is dominated by blue and yellow, giving it a clean, professional look.

KPI ANALYSIS

An isometric illustration of a modern factory floor. The scene is filled with various industrial elements: robotic arms in yellow and blue are positioned over conveyor belts and workstations. Workers in blue uniforms and yellow hard hats are engaged in different tasks, some operating machinery, others handling materials. Stacks of yellow boxes are visible on pallets and in storage areas. The overall color palette is dominated by blue and yellow, creating a clean, industrial aesthetic. A large white rectangular box is superimposed over the center of the image, containing the text 'PRIMARY KPI'S' in bold black letters.

PRIMARY KPI'S

1. List the top 3 and bottom 3 makers for the fiscal years 2023 and 2024 in terms of the number of 2-wheelers sold.

```
5 • WITH top_makers AS
6 ( SELECT
7     m.maker,
8     sum(m.electric_vehicles_sold) AS total_ev_sold
9 FROM sales_by_maker m
10 JOIN dim_date d USING (date)
11 WHERE d.fiscal_year IN (2023,2024) AND m.vehicle_category="2-wheelers"
12 GROUP BY m.maker
13 ORDER BY total_ev_sold desc ),
14
15 ranked AS
16 (SELECT *,
17     DENSE_RANK() OVER(ORDER BY total_ev_sold DESC) AS d,
18     DENSE_RANK() OVER(ORDER BY total_ev_sold ASC) AS d2
19 FROM top_makers)
20 SELECT maker, total_ev_sold
21 FROM ranked
22 WHERE d <= 3 OR d2<=3
23 ORDER BY total_ev_sold DESC
```

Result Grid			Filter Rows:
	maker	total_ev_sold	
▶	OLA ELECTRIC	475072	
	TVS	262836	
	ATHER	184473	
	KINETIC GREEN	9585	
	JITENDRA	8563	
	BATTRE ELECTRIC	4841	

2. Identify the top 5 states with the highest penetration rate in 2-wheeler and 4-wheeler EV sales in FY 2024

```
WITH sales_by_state_fy AS (  
  SELECT  
    s.state,  
    SUM(s.electric_vehicles_sold) AS total_ev_sales,  
    SUM(s.total_vehicles_sold) AS total_vehicle_sales,  
    (SUM(s.electric_vehicles_sold) / SUM(s.total_vehicles_sold)) * 100 AS penetration_rate  
  FROM  
    electric_vehicle_sales_by_state s  
  JOIN  
    dim_date d ON s.date = d.date  
  WHERE  
    d.fiscal_year = '2024'  
  GROUP BY  
    s.state)  
SELECT  
  state,  
  penetration_rate  
FROM  
  sales_by_state_fy  
ORDER BY  
  penetration_rate DESC limit 5;
```

Result Grid		Filter Rows:
	state	penetration_rate
▶	Goa	13.7525
	Kerala	11.5870
	Karnataka	10.1764
	Maharashtra	8.5950
	Delhi	7.7058

3. List the states with negative penetration (decline) in EV sales from 2022 to 2024?

```
WITH sales_by_state_fy AS (  
    SELECT  
        s.state,  
        s.vehicle_category,  
        d.fiscal_year,  
        SUM(s.electric_vehicles_sold) AS total_ev_sales,  
        SUM(s.total_vehicles_sold) AS total_vehicle_sales,  
        (SUM(s.electric_vehicles_sold) / SUM(s.total_vehicles_sold)) * 100 AS penetration_rate  
    FROM  
        electric_vehicle_sales_by_state s  
    JOIN  
        dim_date d ON s.date = d.date  
    WHERE  
        d.fiscal_year IN ('2022', '2023', '2024')  
    GROUP BY  
        s.state, s.vehicle_category, d.fiscal_year  
),  
penetration_change AS (  
    SELECT  
        state,  
        vehicle_category,  
        fiscal_year,  
        penetration_rate,  
        LAG(penetration_rate) OVER (PARTITION BY state, vehicle_category ORDER BY fiscal_year) AS prev_penetration_rate  
    FROM sales_by_state_fy  
)
```

```
SELECT  
    *  
FROM  
    penetration_change  
WHERE  
    penetration_rate < prev_penetration_rate AND fiscal_year = '2024'  
ORDER BY  
    state, vehicle_category;
```

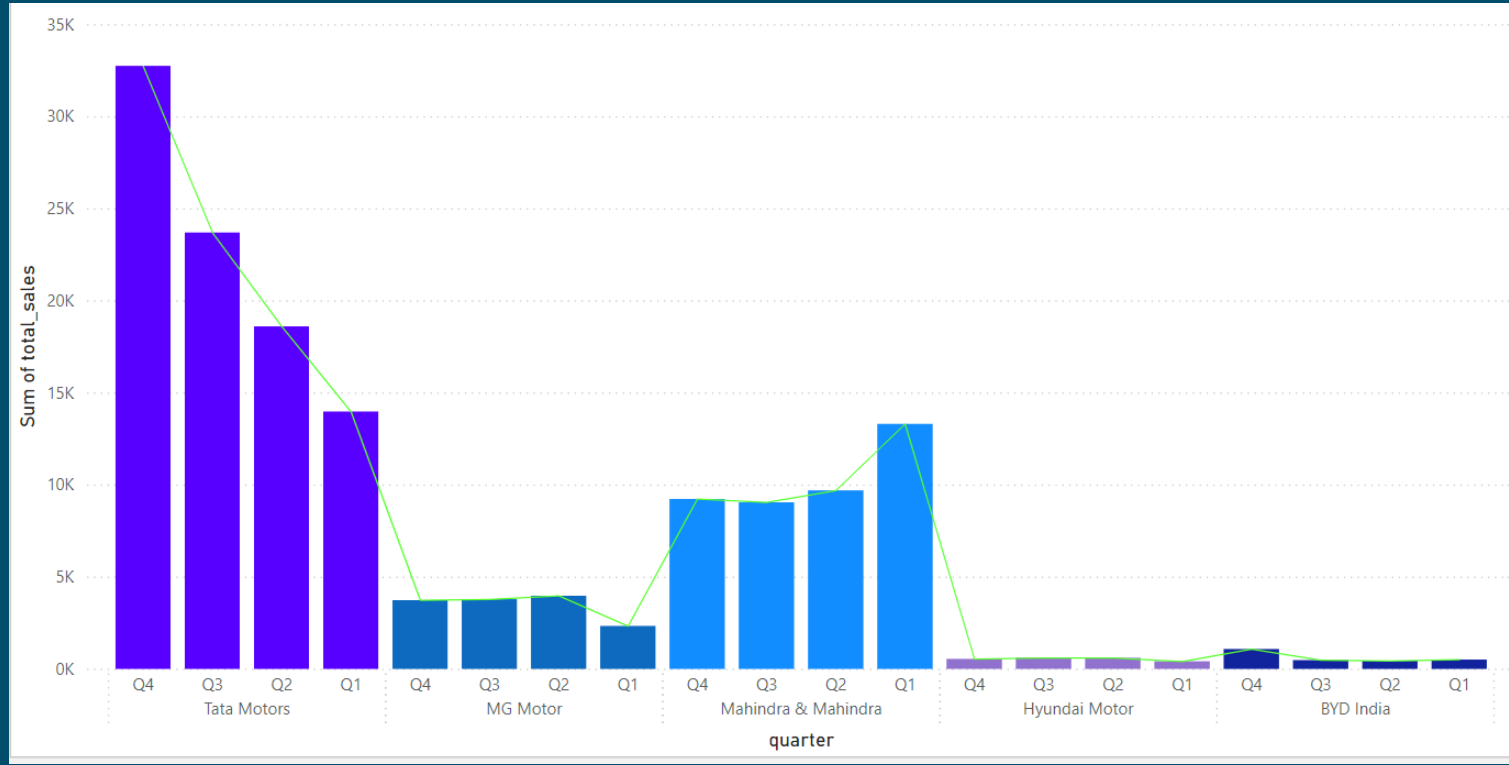
Result Grid	Filter Rows:	Export:	Wrap Cell Content:		
	state	vehide_category	fiscal_year	penetration_rate	prev_penetration_rate
▶	Andhra Pradesh	2-Wheelers	2024	4.5170	4.5827
	Delhi	2-Wheelers	2024	9.4009	9.5236
	Gujarat	2-Wheelers	2024	6.2368	6.7928
	Haryana	2-Wheelers	2024	1.9517	2.7948
	Himachal Pradesh	2-Wheelers	2024	1.3056	1.5895
	Jammu and Kashmir	2-Wheelers	2024	2.5112	2.5319
	Jharkhand	2-Wheelers	2024	1.6802	1.8927
	Ladakh	4-Wheelers	2024	0.0000	0.4288
	Rajasthan	2-Wheelers	2024	5.6002	6.5758
	Uttarakhand	2-Wheelers	2024	3.2998	3.8061

4. What are the quarterly trends based on sales volume for the top 5 EV makers (4-wheelers) from 2022 to 2024?

```
WITH sales_by_maker_quarter AS (  
    SELECT  
        m.make,  
        d.fiscal_year,  
        d.quarter,  
        SUM(m.electric_vehicles_sold) AS total_sales  
    FROM  
        sales_by_maker m  
    JOIN  
        dim_date d ON m.date = d.date  
    WHERE  
        m.vehicle_category = '4-Wheelers' AND  
        d.fiscal_year IN ('2022', '2023', '2024')  
    GROUP BY  
        m.make, d.fiscal_year, d.quarter  
),  
top_5_makers AS (  
    SELECT  
        make  
    FROM  
        sales_by_maker_quarter  
    GROUP BY  
        make  
    ORDER BY  
        SUM(total_sales) DESC  
    LIMIT 5  
)
```

```
SELECT  
    s.make,  
    s.fiscal_year,  
    s.quarter,  
    s.total_sales  
FROM  
    sales_by_maker_quarter s  
JOIN  
    top_5_makers t ON s.make = t.make  
ORDER BY  
    s.make, s.fiscal_year, s.quarter;
```

What are the quarterly trends based on sales volume for the top 5 EV makers (4-wheelers) from 2022 to 2024?



5. How do the EV sales and penetration rates in Delhi compare to Karnataka for 2024?

```
WITH sales_by_state_fy AS (  
    SELECT  
        s.state,  
        SUM(s.electric_vehicles_sold) AS total_ev_sales,  
        SUM(s.total_vehicles_sold) AS total_vehicle_sales,  
        (SUM(s.electric_vehicles_sold) / SUM(s.total_vehicles_sold)) * 100 AS penetration_rate  
    FROM  
        electric_vehicle_sales_by_state s  
    JOIN  
        dim_date d ON s.date = d.date  
    WHERE  
        d.fiscal_year = '2024' AND  
        s.state IN ('Delhi', 'Karnataka')  
    GROUP BY  
        s.state  
)  
SELECT  
    state,  
    total_ev_sales,  
    penetration_rate  
FROM  
    sales_by_state_fy  
ORDER BY  
    penetration_rate desc;
```

	state	total_ev_sales	penetration_rate
▶	Karnataka	160989	10.1764
	Delhi	46724	7.7058

6. List down the compounded annual growth rate (CAGR) in 4-wheeler units for the top 5 makers from 2022 to 2024

```
WITH totalsales AS (  
    SELECT  
        m.maker,  
        d.fiscal_year,  
        SUM(m.electric_vehicles_sold) AS total_sales  
    FROM  
        sales_by_marker m  
    JOIN  
        dim_date d ON m.date = d.date  
    WHERE  
        m.vehicle_category = '4-wheelers' AND  
        d.fiscal_year IN ('2022', '2023', '2024')  
    GROUP BY  
        m.maker, d.fiscal_year),  
  
prev_sales AS (  
    SELECT  
        maker,  
        fiscal_year,  
        total_sales,  
        LAG(total_sales) OVER(PARTITION BY maker ORDER BY fiscal_year) AS prev_yr_sales  
    FROM totalsales  
)
```

```
cagr AS (  
    SELECT  
        maker,  
        (POW(MAX(total_sales) / MIN(prev_yr_sales), 1.0/2) - 1) * 100 AS cagr  
    FROM  
        prev_sales  
    WHERE  
        prev_yr_sales IS NOT NULL  
    GROUP BY  
        maker  
)  
  
SELECT * FROM cagr  
ORDER BY cagr DESC limit 5
```

Result Grid			Filter Rows:
	maker	cagr	
▶	BMW India	1140.9673645990856	
	Volvo Auto India	971.2142642814275	
	BYD India	566.5151342917878	
	Hyundai Motor	255.47663264974256	
	Mercedes-Benz AG	234.5490650412881	

7. List down the top 10 states that had the highest compounded annual growth rate (CAGR) from 2022 to 2024 in total vehicles sold.

```
WITH totalsales AS (  
    SELECT  
        m.state,  
        d.fiscal_year,  
        SUM(m.electric_vehicles_sold) AS total_sales  
    FROM  
        electric_vehicle_sales_by_state m  
    JOIN  
        dim_date d ON m.date = d.date  
    WHERE  
        d.fiscal_year IN ('2022', '2023', '2024')  
    GROUP BY  
        m.state, d.fiscal_year),  
  
prev_sales as (  
    select  
        state,  
        fiscal_year,  
        total_sales,  
        lag(total_sales) over(partition by state order by fiscal_year) as prev_yr_sales  
    from totalsales  
),
```

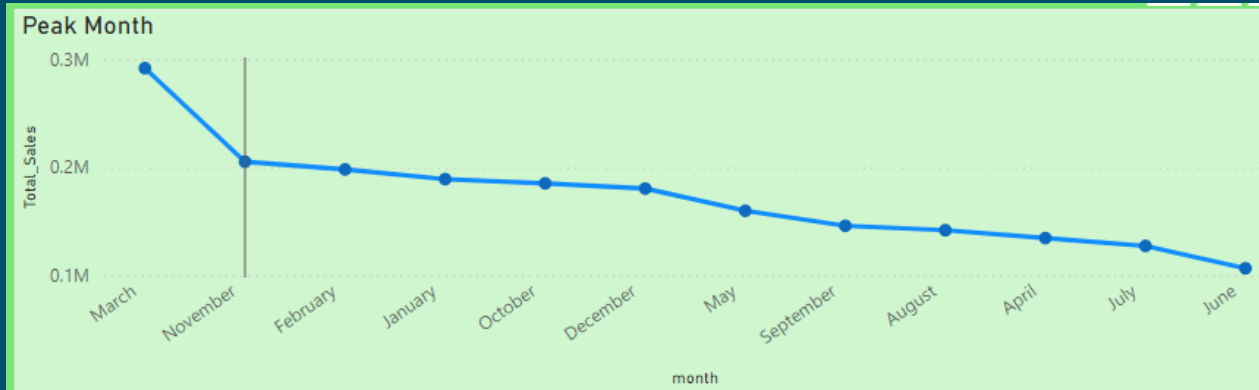
```
cagr as (  
    SELECT  
        state,  
        (POW(MAX(total_sales) / MIN(prev_yr_sales), 1/2) - 1) * 100 AS cagr  
    FROM  
        prev_sales  
    WHERE  
        prev_yr_sales IS NOT NULL  
    GROUP BY  
        state  
    )  
select * from cagr  
order by cagr desc limit 10
```

	state	cagr
▶	Meghalaya	476.6281297335398
	Tripura	229.5017884169978
	Nagaland	200
	Chandigarh	164.57513110645908
	Chhattisgarh	150.891660861815
	West Bengal	150.61562932107807
	Goa	146.44833718246102
	Manipur	143.31050121192877
	DNH and DD	137.84749014862444
	Uttar Pradesh	137.7049003281169

8. What are the peak and low season months for EV sales based on the data from 2022 to 2024?

```
with cte as
(
  select
    monthname(d.date) as month,
    sum(electric_vehicles_sold) as total_sales
  from electric_vehicle_sales_by_state
  join dim_date d using (date)
  where
    d.fiscal_year in (2022,2023,2024)
  group by month
  order by total_sales desc)

select *,
       dense_rank() over(order by total_sales desc) as peak_month
from cte
```



9. What is the projected number of EV sales (including 2-wheelers and 4-wheelers) for the top 10 states by penetration rate in 2030, based on the compounded annual growth rate (CAGR) from previous years?

```
WITH recent_sales AS (  
    -- total sales & penetration_rate from latest year (2024)  
    SELECT  
        s.state,  
        SUM(s.electric_vehicles_sold) AS ev_sales_2024,  
        SUM(s.total_vehicles_sold) AS total_sales_2024,  
        (SUM(s.electric_vehicles_sold) / SUM(s.total_vehicles_sold)) * 100 AS penetration_rate_2024  
    FROM  
        electric_vehicle_sales_by_state s  
    JOIN  
        dim_date d ON s.date = d.date  
    WHERE  
        d.fiscal_year = '2024'  
    GROUP BY  
        s.state  
) ,  
top_10_states AS (  
    -- top 10 states by penetration rate in 2024  
    SELECT  
        state  
    FROM  
        recent_sales  
    ORDER BY  
        penetration_rate_2024 DESC  
    LIMIT 10  
) ,
```

What is the projected number of EV sales (including 2-wheelers and 4- wheelers) for the top 10 states by penetration rate in 2030, based on the compounded annual growth rate (CAGR) from previous years?

```
cagr_calculation AS (  
  -- CAGR calculation from FY 2022 to FY 2024 for the top 10 states  
  SELECT  
    s.state,  
    SUM(CASE WHEN d.fiscal_year = '2022' THEN s.electric_vehicles_sold ELSE 0 END) AS ev_sales_2022,  
    SUM(CASE WHEN d.fiscal_year = '2024' THEN s.electric_vehicles_sold ELSE 0 END) AS ev_sales_2024,  
    POWER((SUM(CASE WHEN d.fiscal_year = '2024' THEN s.electric_vehicles_sold ELSE 0 END) /  
           SUM(CASE WHEN d.fiscal_year = '2022' THEN s.electric_vehicles_sold ELSE 0 END)), 1.0/2) - 1 AS cagr  
  FROM  
    electric_vehicle_sales_by_state s  
  JOIN  
    dim_date d ON s.date = d.date  
  WHERE  
    d.fiscal_year IN ('2022', '2024')  
    AND s.state IN (SELECT state FROM top_10_states)  
  GROUP BY  
    s.state  
)
```

```
projected_sales AS (  
  -- Project the EV sales for 2030 using the CAGR  
  SELECT  
    state, ev_sales_2024,  
    ev_sales_2024 * POWER(1 + cagr, 2030 - 2024) AS projected_ev_sales_2030  
  FROM  
    cagr_calculation  
)  
  
-- projected EV sales for 2030  
SELECT  
  state, ev_sales_2024,  
  round(projected_ev_sales_2030/ 1000000,2) as proj_sales_in_mill  
FROM  
  projected_sales  
ORDER BY  
  projected_ev_sales_2030 DESC;
```

	state	ev_sales_2024	proj_sales_in_mill
▶	Maharashtra	197169	13.35
	Kerala	73938	11.78
	Karnataka	160989	8.38
	Chhattisgarh	28540	7.12
	Odisha	39118	2.73
	Goa	10799	2.42
	Tamil Nadu	94314	1.58
	Delhi	46724	1.05
	Chandigarh	2877	0.99
	Puducherry	3098	0.23

10. Estimate the revenue growth rate of 4-wheeler and 2-wheelers EVs in India for 2022 vs 2024 and 2023 vs 2024, assuming an average unit price

Vehicle_category	Average Price
2-Wheelers	₹ 85,000.00
4-Wheelers	₹ 15,00,000.00

```
WITH sales_revenue AS (  
    -- Step 1: Calculate total revenue for each vehicle category and year  
    SELECT  
        d.fiscal_year,  
        s.vehicle_category,  
        SUM(s.electric_vehicles_sold *  
            CASE  
                WHEN s.vehicle_category = '2-Wheelers' THEN 85000  
                WHEN s.vehicle_category = '4-Wheelers' THEN 1500000  
            END) AS total_revenue  
    FROM  
        electric_vehicle_sales_by_state s  
    JOIN  
        dim_date d ON s.date = d.date  
    WHERE  
        d.fiscal_year IN ('2022', '2023', '2024')  
    GROUP BY  
        d.fiscal_year, s.vehicle_category
```

Estimate the revenue growth rate of 4-wheeler and 2-wheelers EVs in India for 2022 vs 2024 and 2023 vs 2024, assuming an average unit price

Vehicle_category	Average Price
2-Wheelers	₹ 85,000.00
4-Wheelers	₹ 15,00,000.00

```
growth_rate AS (  
    -- Step 2: Calculate the revenue growth rate between the years  
    SELECT  
        vehicle_category,  
        (MAX(CASE WHEN fiscal_year = '2024' THEN total_revenue ELSE 0 END) /  
         MAX(CASE WHEN fiscal_year = '2022' THEN total_revenue ELSE 0 END) - 1) * 100 AS growth_2022_2024,  
        (MAX(CASE WHEN fiscal_year = '2024' THEN total_revenue ELSE 0 END) /  
         MAX(CASE WHEN fiscal_year = '2023' THEN total_revenue ELSE 0 END) - 1) * 100 AS growth_2023_2024  
    FROM  
        sales_revenue  
    GROUP BY  
        vehicle_category  
)  
  
-- Final Step: Display the growth rates  
SELECT  
    vehicle_category,  
    growth_2022_2024,  
    growth_2023_2024  
FROM  
    growth_rate;  
/*
```

STORED PROCEDURE

In the first question of KPI's we list down the top & bottom 3 makes for 2023 -24

But what if we need the top & bottom 5 makes for 2022 -24?

Do we need to rewrite the code again ?

As criteria changes rewriting code will be a time taking task and not recommended

So we need to create a Stored procedure where we declare values that we need to provide as variables and every time with just few inputs we can alter the output.

Follow the example code of first stored procedure

STORED PROCEDURE

```
CREATE DEFINER='root'@'localhost' PROCEDURE `CAGR_by_maker`(  
  in_fiscal_year TEXT,  
  in_vehicle_category VARCHAR(45),  
  in_top_n int  
)  
BEGIN  
  WITH totalsales AS (  
    SELECT  
      m.make,  
      d.fiscal_year,  
      SUM(m.electric_vehicles_sold) AS total_sales  
    FROM  
      sales_by_maker m  
    JOIN  
      dim_date d ON m.date = d.date  
    WHERE  
      m.vehicle_category = in_vehicle_category AND find_in_set(d.fiscal_year , in_fiscal_year)  
    GROUP BY  
      m.make, d.fiscal_year),
```

Variables

Using created Variable in the code

STORED PROCEDURE

OUTPUT

```
prev_sales AS (  
  SELECT  
    maker,  
    fiscal_year,  
    total_sales,  
    LAG(total_sales) OVER(PARTITION BY maker ORDER BY fiscal_year) AS prev_yr_sales  
  FROM totalsales  
,  
cagr AS (  
  SELECT  
    maker,  
    (POW(MAX(total_sales) / MIN(prev_yr_sales), 1.0/2) - 1) * 100 AS cagr  
  FROM  
    prev_sales  
  WHERE  
    prev_yr_sales IS NOT NULL  
  GROUP BY  
    maker  
)  
SELECT * FROM cagr  
ORDER BY cagr DESC limit in_top_n;  
END
```

```
1 • call ev.top_bott_makers('2-wheelers', '2022', '2024', 5);  
2
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [A](#)

maker	total_ev_sold
OLA ELECTRIC	489473
TVS	272575
ATHER	204449
HERO ELECTRIC	170394
AMPERE	167274
BEING	18392
BGAUSS	15051
JITENDRA	12438
KINETIC GREEN	9585
BATTRE ELECTRIC	4841

An isometric illustration of a modern factory floor. The scene is filled with various industrial elements: robotic arms (yellow and black) are positioned over conveyor belts and workstations, moving orange boxes. Workers in blue uniforms and yellow hard hats are engaged in different tasks, some operating machinery, others handling materials. A blue forklift is visible on the right side. The floor is marked with yellow and black safety lines. The overall color palette is dominated by blue, yellow, and orange.

SECONDARY KPI'S

What are the primary reasons for customers choosing 4-wheeler EVs in 2023 and 2024

1. Cost Savings

- Fuel Savings
- Maintenance Savings.
- Total Cost of Ownership (TCO)



2. Environmental Concerns

- Reduction in Carbon Footprint
- Public Awareness



3. Government Incentives

- Subsidies and Tax Breaks
- Reduced Registration Fees
- Charging Infrastructure Development



4. Social and life style factors

- Status Symbol
- Quiet Operation



How do government incentives and subsidies impact the adoption rates of 2-wheelers and 4-wheelers? Which states in India provided most subsidies?



Impact on Adoption Rates

- Increased Affordability
- Demand Stimulation
- Infrastructure Development



Top states providing subsidies

- Maharashtra
- Gujarat
- Delhi
- Tamil Nadu
- Karnataka

How does the availability of charging stations infrastructure correlate with the EV sales and penetration rates in the top 5 states?

After analysis I observe There's a strong positive correlation between the availability of charging stations infrastructure and EV sales and penetration rates in India's top 5 states this is due to ,

- Convenience and Accessibility
- Government Incentives



State Name	No. of Operational PCS	state	penetration_rate
Maharashtra	3079	Goa	9.84
Delhi	1886	Karnataka	7.84
Karnataka	1041	Delhi	6.76
Kerala	852	Kerala	6.64
Tamil Nadu	643	Maharashtra	6.49

Which state of India is ideal to start the manufacturing unit? (Based on subsidies provided, ease of doing business, stability in governance etc.)

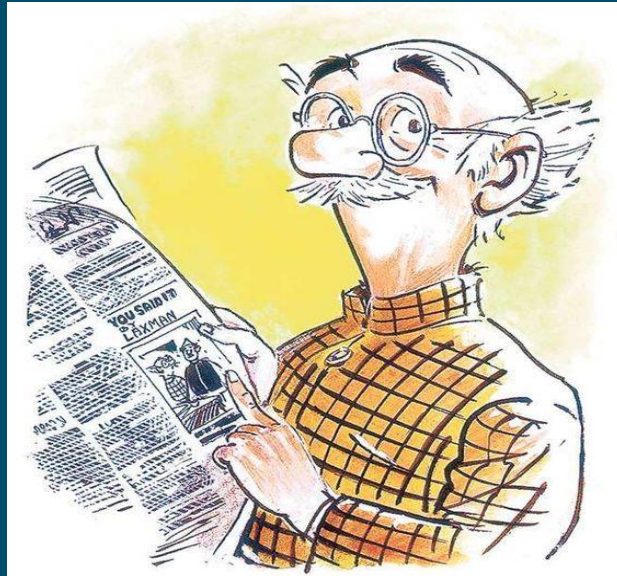
The decision of where to establish an EV manufacturing unit in India depends on several factors, including subsidies, ease of doing business, stability in governance, infrastructure, and market potential. While there is no one-size-fits-all answer, here are some states that are generally considered favorable for EV manufacturing:



- Maharashtra
- Gujarat
- Tamil Nadu
- Karnataka
- Andhra Pradesh

But if we have to choose one Maharashtra is the favorable state giving highest subsidies and having large market supports to set up EV plant

. Who should be the brand ambassador if AtliQ Motors launches their EV/Hybrid vehicles in India and why?



THE COMMON MAN

RK Laxman's cartoons | The Times of India

Your top 3 recommendations for AtliQ Motors

- Choose Maharashtra as the Primary Location
- Focus on making budget EV's and Charging Infrastructure Development
- Leverage Government Policies and Incentives



An isometric illustration of a modern factory floor. The scene is filled with various industrial elements: robotic arms in yellow and blue, workers in blue uniforms and hard hats, conveyor belts, and stacks of yellow boxes. The background shows a complex network of pipes and structural elements. The overall color palette is dominated by blue and yellow, giving it a clean, industrial feel.

VISUALIZATION

MySQL connection to Power BI

fx

= MySQL.Database("127.0.0.1:3306", "ev", [ReturnSingleDatabase=true, Query="call ev.top_bott_makers

ABC maker

1.2 total_ev_sold

Valid

100%

Error

0%

Empty

0%

6 distinct, 6 unique

Valid

100%

Error

0%

Empty

0%

6 distinct, 6 unique

1	OLA ELECTRIC	475072
2	TVS	262836
3	ATHER	184473
4	KINETIC GREEN	9585
5	JITENDRA	8563
6	BATTRE ELECTRIC	4841

MySQL database

Server

127.0.0.1:3306

Database

ev

Advanced options

Command timeout in minutes (optional)

SQL statement (optional, requires database)

call ev.top_bott_makers('2-wheelers', '2023', '2024', 3);

☒ Include relationship columns

☐ Navigate using full hierarchy

An isometric illustration of a modern factory floor. The scene is filled with various industrial elements: robotic arms in yellow and blue, workers in blue uniforms and white lab coats, conveyor belts, and stacks of yellow boxes. The background is a solid blue color, and the overall style is clean and professional.

KPI BY STATE



2022

2023

2024

2-Wheelers

4-Wheelers

Q1

Q2

Q3

Q4

maker

All

2M

electric_vehicles_sold

57M

total_vehicles_sold

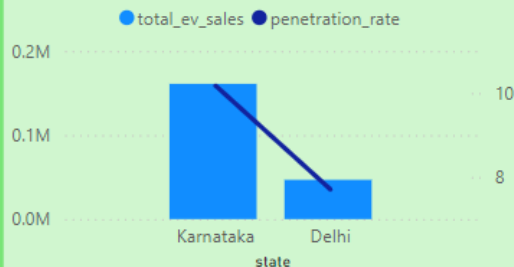
3.61

penetration_rate

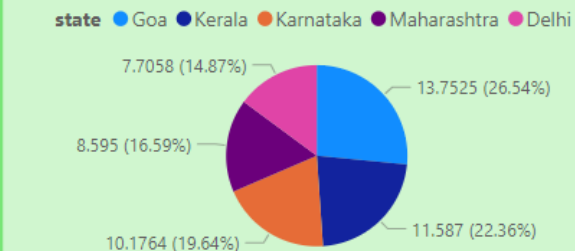
93.91

CAGR

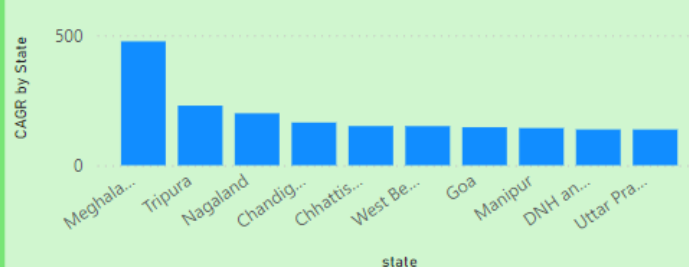
Delhi vs Karnataka penetration_rate in 2024



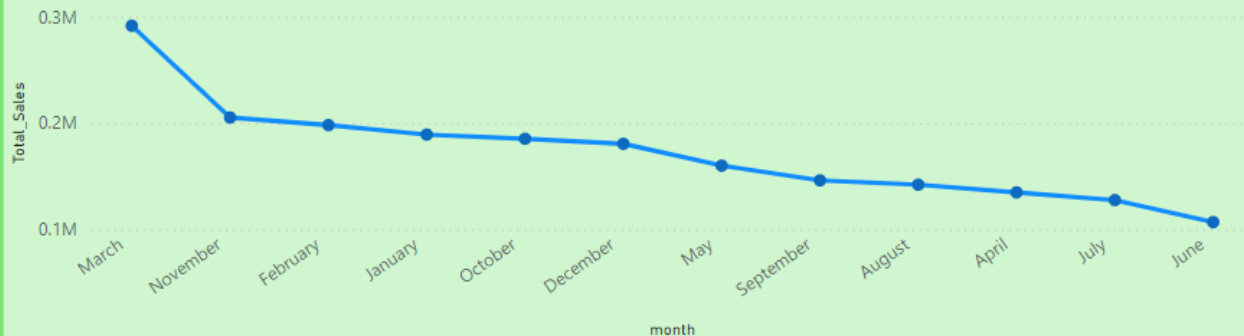
Penetration_rate for Top 5 states by state



CAGR by State by state



Peak Month



Projected ev sales by 2030 vs current year (2024)

state	Ev_sales_2024	Proj_sales_in_mill
Maharashtra	1,97,169.00	13.35
Karnataka	1,60,989.00	8.38
Tamil Nadu	94,314.00	1.58
Kerala	73,938.00	11.78
Delhi	46,724.00	1.05
Odisha	39,118.00	2.73
Chhattisgarh	28,540.00	7.12
Goa	10,799.00	2.42
Puducherry	3,098.00	0.23
Chandigarh	2,877.00	0.99
Total	6,57,566.00	49.63

An isometric illustration of a modern factory floor. The scene is filled with various industrial elements: robotic arms (yellow and black) are positioned over conveyor belts and workstations, moving orange boxes. Workers in blue uniforms and yellow hard hats are engaged in different tasks, some operating machinery, others handling materials. A large blue forklift is visible on the right. The floor is marked with yellow and black safety lines. The overall color palette is dominated by blue, yellow, and orange.

KPI BY MAKER



2022

2023

2024

2-Wheelers

4-Wheelers

Q1

Q2

Q3

Q4

maker

All

2M

Electric vehicles sold

57M

total vehicles sold

3.61

penetration rate

93.91

CAGR

Top_3 makers in 2-wheelers market

maker ● OLA ELECTRIC ● TVS ● ATHER

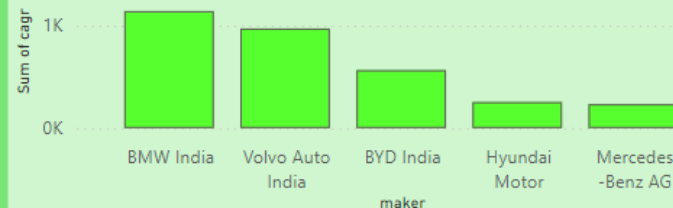


Bottom_3 makers in 2-wheelers market

maker ● KINETIC GREEN ● JITENDRA ● BATTRE ELECTRIC

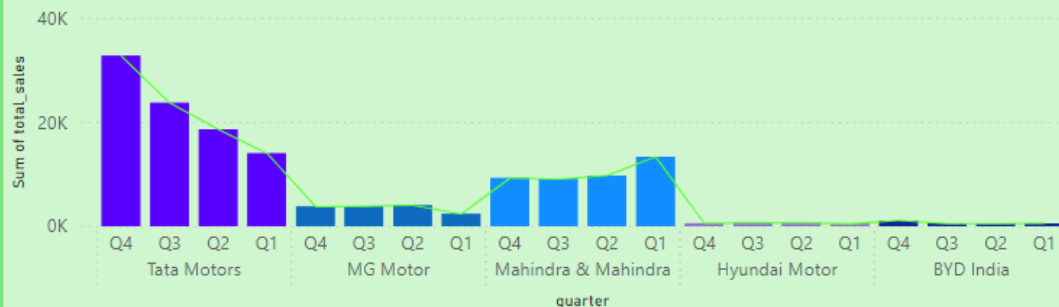


CAGR by maker



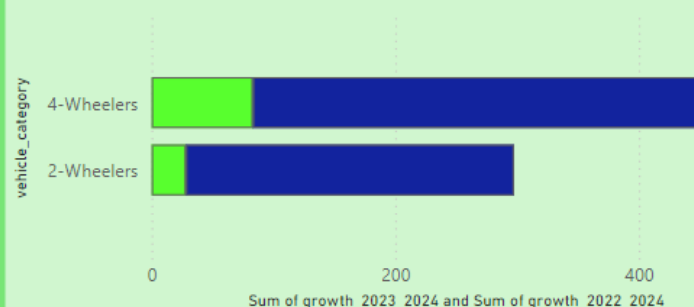
Quarterly sales trends for TOP 5 makers

● Sum of total_sales ● Sum of total_sales



Growth rate 2022/2024 vs 2023/2024 by vehicle category

● Sum of growth_2023_2024 ● Sum of growth_2022_2024





Total Number of charging stations by state

State Name No. of Operational PCS

Maharashtra	3079
Delhi	1886
Karnataka	1041
Kerala	852
Tamil Nadu	643
Uttar Pradesh	582
Rajasthan	500
Telangana	481
Gujarat	476
Haryana	377
Madhya Pradesh	341
Andhra Pradesh	327
West Bengal	318
Odisha	198
Punjab	158
Chhattisgarh	149
Jharkhand	135
Bihar	124
Goa	113
Assam	86
Uttarakhand	76
Jammu & Kashmir	47
Himachal Pradesh	44
Pondicherry	23
Meghalaya	21
Tripura	18
Manipur	17
Chandigarh	12
Arunachal Pradesh	9
Mizoram	6

Fiscal Year:

The fiscal year is a one-year period used for financial reporting and budgeting, starting on April 1st and ending on March 31st of the following year in India.

Penetration Rate:

This metric represents the percentage of total vehicles that are electric within a specific region or category.

It is calculated as:

$$\text{Penetration Rate} = (\text{Electric Vehicles Sold} / \text{Total Vehicles Sold}) * 100$$

This indicates the adoption level of electric vehicles.

CAGR = Compound Annual growth rate :

Compound Annual Growth Rate (CAGR): CAGR measures the mean annual growth rate over a specified period longer than one year.

It is calculated as:

$$\text{CAGR} = [(\text{Ending Value} / \text{Beginning Value})^{** 1/n}] - 1$$

Amaresam Sai Bharath Chand



Checkout Project Description in
Git hub



My linkedin profile



Checkout Project Explanation
Video

CREDITS

The dataset is taken from the Vahan Sewa.

Thanks to the Vahan Sewa for providing datasets for public access which is a great learning asset – feel free to explore them here: [VAHAN SEVA](#)

This project is part of code basics [resume challenge 12](#)

Big thanks to **Dhaval Patel Sir, Hemanand Vadivel Sir**, and the entire **Codebasics** team for their invaluable guidance throughout this journey!

